



TRAFFIC CONGESTION AND VEHICLE EMISSIONS

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1. ABSTRACT

Traffic congestion and vehicle emissions are major problems in urban areas. The increasing use of private cars and two-wheelers leads to a greater number of vehicles on roads, resulting in congestion, increased travel time, fuel consumption and air pollution. Transportation is an important source of air pollution and greenhouse-gas emissions. Vehicle-related pollution can come from exhaust as well as non-exhaust sources such as brake and tyre wear and road dust.

One effective approach to addressing this problem is to encourage people to use public transportation instead of individual private vehicles. Buses, trains and metro systems can carry many passengers in a single journey and can therefore reduce the number of individual vehicles required for passenger travel.

The proposed solution is a Public Transport Recommendation System (PTRS). It is an IT-based system designed to recommend suitable public transportation options to users based on their source, destination, travel time, cost and other preferences. The main purpose of the system is not simply to provide directions but to encourage users to shift from private vehicles to public transport.

2. INTRODUCTION

Transportation plays an important role in everyday life. People travel to colleges, schools, offices, hospitals, shopping centres and other destinations. As cities grow and populations increase, the demand for transportation also increases.

One major consequence of increasing transportation demand is the growth in the number of private vehicles. Cars and motorcycles provide convenience and flexibility, but excessive dependence on individual vehicles can contribute to traffic congestion and environmental pollution.

Traffic congestion occurs when the number of vehicles using a road becomes too high for the available road capacity. During congested conditions, vehicles may spend considerable amounts of time waiting or moving slowly. Frequent stopping and starting can increase fuel consumption and contribute to emissions.

Vehicle emissions are an important environmental concern. Transport-related air pollution can affect air quality and human health. The World Health Organization identifies transport as an important source of air pollution and greenhouse-gas emissions and describes health risks associated with transport-related air pollution.

Public transportation provides an alternative to individual vehicle travel. A bus, train or metro can carry multiple passengers during the same journey. Therefore, increasing public transport usage can potentially reduce the number of individual private vehicles on roads.

However, the availability of public transportation does not always mean that people will use it. Many users may prefer private vehicles because they consider them more convenient or because they do not know the available public transportation routes.

Some common difficulties include:

- Not knowing which bus or train to use.
- Difficulty understanding routes.
- Uncertainty about travel time.
- Lack of information about fares.
- Difficulty comparing transportation options.
- Lack of awareness about environmental benefits.
- Preference for private vehicle convenience.

3. PROBLEM STATEMENT

3.1 Problem Definition

Excessive dependence on private vehicles contributes to traffic congestion and vehicle emissions, while users may lack convenient information and motivation to choose public transportation.

Although public transport may be available, users may not know which service is most suitable for their journey.

3.2 Causes of the Problem

1. Increasing Private Vehicle Usage.
2. Convenience of Private Vehicles.
3. Lack of Public Transport Awareness.
4. Difficulty in Comparing Options.
5. Limited Environmental Awareness.

3.3 Effects of the Problem

6. Longer travel times.
7. Increased fuel consumption.
8. Increased transportation costs.
9. Air pollution.
10. Greenhouse-gas emissions.
11. Noise pollution.
12. Reduced productivity.
13. Greater pressure on urban infrastructure.

WHO highlights sustainable transportation as an important approach for reducing risks associated with vehicular air pollution while providing environmental and climate benefits.

3.4 Need for the Proposed Solution

There is a need for a simple system that makes public transportation easier to choose.

The proposed system focuses on the following question:

Q. How can an IT system encourage people to replace unnecessary private vehicle journeys with public transportation?

The answer is to provide users with personalised public transport recommendations at the time they are planning their journey.

4. SUSTAINABILITY ANALYSIS

4.1 Sustainable Development Goals

The Public Transport Recommendation System supports several Sustainable Development Goals (SDGs).

SDG 11 – Sustainable Cities and Communities

It focuses on providing safe, affordable, accessible and sustainable transport systems and specifically calls for improved public transportation.

SDG 3 – Good Health and Well-being

Vehicle-related air pollution can have negative effects on human health. WHO identifies transport-related air pollution as a health concern. By encouraging sustainable transportation, the proposed system can contribute indirectly to healthier urban environments.

SDG 9 – Industry, Innovation and Infrastructure

The project uses information technology to provide an innovative solution to an urban transportation problem. Instead of relying only on physical infrastructure development, the system uses software and data to improve transportation decision-making.

SDG 13 – Climate Action

Transportation contributes to greenhouse-gas emissions. Encouraging people to choose public transportation instead of individual private vehicles can potentially reduce emissions associated with passenger travel.

4.2 ISO 14001

ISO 14001:2026 is the international standard for Environmental Management Systems. It provides a framework for organisations to manage environmental responsibilities systematically and improve environmental performance. The Public Transport Recommendation System is not being claimed as an ISO-certified product. However, the principles of ISO 14001 can be applied while designing and evaluating the project.

Environmental Objective- To encourage users to replace suitable private vehicle journeys with public transportation and thereby contribute to sustainable mobility.

Environmental Aspects

1. Vehicle emissions.
2. Fuel consumption.
3. Private vehicle usage.

Monitoring

1. Number of searches.
2. Number of public transport recommendations.
3. Number of users selecting public transportation.

Continual Improvement

1. Updating transportation information.

2. Improving recommendations.
3. Updating emission factors.

4.3 Life Cycle Assessments

Life Cycle Assessment (LCA) is used to study the environmental impact of our Public Transport Recommendation System throughout its entire life cycle. Even though our project is software-based, it still uses resources and energy during different stages.

During the development stage, computers, electricity and internet services are required to create and test the system. During the deployment and usage stage, the website needs servers or cloud services to store data and process user requests, which consume electricity. The maintenance stage includes updating the software, database and public transport information, which also requires computing resources.

By recommending public transportation and showing its advantages compared with private vehicles, the system can encourage users to choose buses, trains or metro services.

5. PROPOSED ENGINEERING/IT SOLUTION

5.1 Overview

The proposed solution is a: **PUBLIC TRANSPORT RECOMMENDATION SYSTEM**

It is a web-based application designed to recommend suitable public transportation options to users.

The main objective is: To shift users from private vehicles to public transport by making public transportation easier to discover, compare and choose.

The system can recommend transportation such as:

- Bus.
- Train.
- Metro.
- Public shuttle services.

5.2 Objectives

1. To encourage users to use public transportation.
2. To reduce dependence on private vehicles.
3. To provide suitable public transport recommendations.
4. To make public transport information easier to understand.
5. To provide estimated travel time.
6. To provide estimated transportation cost.
7. To communicate environmental benefits.
8. To promote sustainable travel behaviour.
9. To contribute to reducing traffic congestion.
10. To demonstrate how IT can address an environmental problem.

5.3 Target users

1. Students- Students can find affordable public transportation for travelling between home and college.
2. Employees- Employees can use the system to find public transport for daily commuting.
3. Daily Commuters- Regular commuters can compare available public transportation routes.
4. Visitors- Visitors who are unfamiliar with an area can use the system to understand public transportation.
5. Environmentally Conscious Users- Users who want to make environmentally responsible travel choices can use the system to identify public transport options.

5.6 Major Features

1. Source and Destination Input- The user enters where they are travelling from and where they want to go.
2. Public Transport Search- The system identifies available public transportation options.

3. Public Transport Recommendation- The system selects a suitable option based on user preferences.
4. Travel-Time Information- The system displays estimated travel duration.
5. Cost Information- The system displays the estimated fare.
6. Recommendation Explanation- The system explains why a particular option was selected.
7. Sustainability Message- The website can display messages such as:

6. FEASIBILITY ANALYSIS

6.1 Technical Feasibility

The system is technically feasible because it can be developed using commonly available technologies.

Possible requirements include:

- Computer.
- Internet connection.
- Web browser.
- Web server.
- Database.

6.2 Economic Feasibility

The proposed system is relatively inexpensive to develop as a prototype. Open-source technologies can be used to reduce development costs. Potential costs in a real-world implementation include:

- Website hosting.
- Server infrastructure.
- Database services.
- Transportation data services.
- Mapping services.
- Maintenance.
- Technical support.

6.3 Social Feasibility

The system can benefit:

- Students.
- Employees.
- Daily commuters.
- Visitors.
- Public transportation users.

The system should also be designed to be accessible and easy to understand.

7. EXPECTED ENVIRONMENTAL BENEFITS

7.1 Reduction in Private Vehicle Usage

The primary goal of the project is to encourage people to replace suitable private vehicle journeys with public transportation. If users follow the recommendations, the number of individual vehicles on roads could potentially decrease.

7.2 Reduction in Traffic Congestion

Shifting passengers from individual cars to buses, trains or metro systems can potentially reduce the number of vehicles required for passenger movement.

7.3 Reduction in Fuel Consumption

Fewer private vehicle journeys can result in lower fuel consumption. Public transportation allows many passengers to share a journey rather than requiring each passenger to travel in an individual vehicle.

7.4 Reduction in Vehicle Emissions

The system encourages users to select public transportation and provides environmental information to support their decision. Transport is an important source of air pollution and greenhouse-gas emissions, making transportation choices relevant to environmental sustainability.

7.5 Improved Urban Air Quality

A reduction in vehicle activity can potentially reduce transportation-related pollution. This can contribute to healthier urban environments.

7.6 Increased Environmental Awareness

The system can help users understand that transportation choices have environmental consequences.

7.7 Promotion of Sustainable Cities

The system directly promotes public transportation, which is an important component of sustainable urban mobility.

8. JUSTIFICATION OF THE PROPOSED SOLUTION

The Public Transport Recommendation System is a suitable solution because it directly targets the relationship between private vehicle dependence, congestion and emissions.

8.1 Directly Addresses the Problem

Problem:- Large numbers of individual private vehicles.

Proposed solution:- Recommend public transportation.

Expected result:- Potential shift from individual private vehicles to shared transportation.

8.2 Makes Existing Public Transport Easier to Use

The system does not necessarily require new transportation infrastructure. Instead, it helps users understand and use existing public transport services.

8.3 Encourages Behavioural Change

The system provides information at the exact point when the user is deciding how to travel. This makes it possible to influence the transportation choice.

8.4 Environmentally Focused

Unlike a system that only focuses on speed, the proposed system includes environmental impact as an important factor.

8.5 Affordable and Scalable

The system can begin as a simple website and later expand to include:

- More routes.
- More cities.
- Real-time data.
- AI.
- Smart-city integration.

8.6 Supports Health and Climate Objectives

Sustainable transportation can provide health and environmental benefits, while transport-related air pollution is a recognised health concern.

Therefore, the project addresses not only traffic congestion but also broader sustainability concerns.

9. CONCLUSION

Traffic congestion and vehicle emissions are significant problems in modern urban areas. Excessive dependence on private vehicles contributes to crowded roads, increased fuel consumption and transportation-related pollution.

The Public Transport Recommendation System proposed in this report provides an IT-based solution focused on encouraging people to shift from private vehicles to public transportation.

The system accepts information such as the user's source and destination and provides suitable public transportation recommendations. It can compare travel time, cost, convenience and estimated environmental impact, allowing users to make more informed transportation decisions.

The key feature of the proposed system is its public-transport-first recommendation approach. Instead of simply showing transportation information, it actively encourages users to consider public transportation as an alternative to individual private vehicle travel.

The proposed system is technically, economically and operationally feasible as a software prototype. It can be developed using standard web technologies and can later be expanded with real-time data, artificial intelligence and smart-city integration.

However, the system alone cannot solve traffic congestion or vehicle emissions. Its effectiveness depends on the availability and quality of public transportation and whether users actually change their travel behaviour.

Overall, the Public Transport Recommendation System demonstrates how information technology can be used to promote sustainable transportation. By making public transport easier to discover, understand and choose, the system can contribute to reducing dependence on private vehicles and support the development of cleaner and more sustainable cities.

10. REFERENCES

1. United Nations – Sustainable Development Goal 11: Sustainable Cities and Communities. Target 11.2 addresses safe, affordable, accessible and sustainable transport systems and expanding public transportation.
United Nations – Goal 11
2. World Health Organization – Transport: Sectoral Solutions for Air Pollution and Health. Provides information about transport-related air pollution, emissions and associated health concerns.
WHO – Transport and Air Pollution
3. World Health Organization – Sustainable Transport for Health. Discusses health, environmental and climate benefits associated with sustainable transport.
WHO – Sustainable Transport for Health

4. International Organization for Standardization – ISO 14001:2026. Environmental management systems and requirements for systematic environmental management and improvement.
ISO 14001:2026
5. World Health Organization – Strategies for Healthy and Sustainable Transport. Discusses sustainable transportation approaches, including public transport.
WHO – Healthy and Sustainable Transport